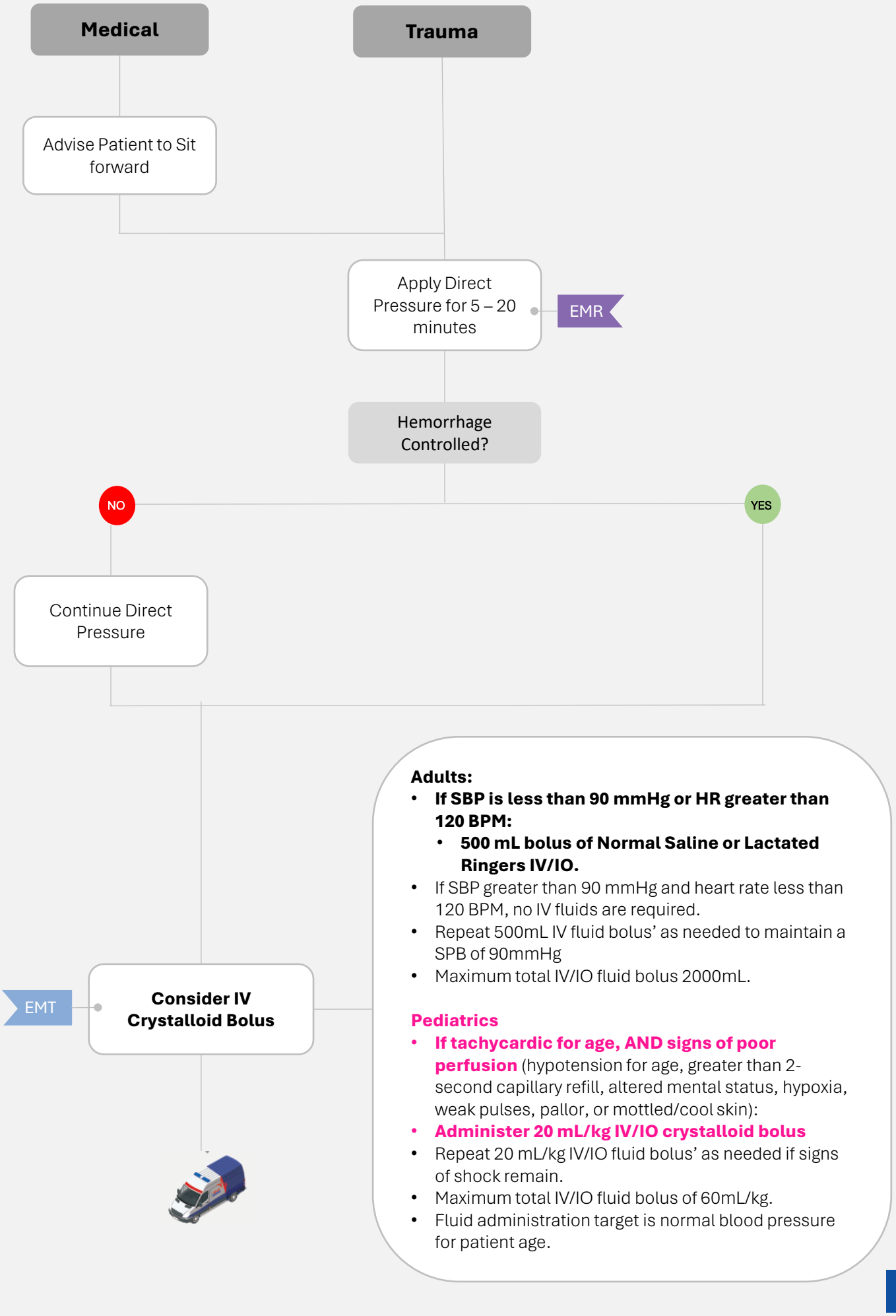
Medical
Trauma
Advise Patient to Sit forward
Apply Direct Pressure for 5 – 20 minutes
EMR
Hemorrhage Controlled?
NO
YES
Continue Direct Pressure
EMT
Consider IV Crystalloid Bolus
Adults:
• If SBP is less than 90 mmHg or HR greater than 120 BPM:
• 500 mL bolus of Normal Saline or Lactated Ringers IV/IO.
• If SBP greater than 90 mmHg and heart rate less than 120 BPM, no IV fluids are required.
• Repeat 500mL IV fluid bolus' as needed to maintain a SPB of 90mmHg
• Maximum total IV/IO fluid bolus 2000mL.
Pediatrics
• If tachycardic for age, AND signs of poor perfusion (hypotension for age, greater than 2-second capillary refill, altered mental status, hypoxia, weak pulses, pallor, or mottled/cool skin):
• Administer 20 mL/kg IV/IO crystalloid bolus
• Repeat 20 mL/kg IV/IO fluid bolus' as needed if signs of shock remain.
• Maximum total IV/IO fluid bolus of 60mL/kg.
• Fluid administration target is normal blood pressure for patient age.

Patient Care Goals

1. Control acute epistaxis to prevent airway compromise or significant blood loss.

Patient Inclusion & Exclusion Criteria

Inclusion Criteria

- Patients presenting with epistaxis (nosebleed), including:
  - Active nasal bleeding, unilateral or bilateral.
  - Symptoms ranging from mild, self-limiting bleeding to severe, persistent hemorrhage.
  - Associated symptoms such as dizziness, weakness, or pallor due to blood loss.
- Suspected underlying causes such as hypertension, trauma, coagulopathy, or medication use (e.g., anticoagulants).

Exclusion Criteria

- Patients with no active bleeding or resolved epistaxis prior to arrival.

Key Patient Assessment Points

1. **Patient History**
   - Determine onset, duration, and frequency of bleeding (e.g., first episode or recurrent).
   - Identify potential triggers (e.g., nasal irritation, trauma, recent surgery, or dry environment).
   - Inquire about medication use (e.g., anticoagulants, antiplatelets, or nasal sprays).
   - Confirm history of bleeding disorders, hypertension, or recent nasal procedures.
2. **Bleeding**
   - Assess the severity of bleeding (e.g., dripping, streaming, or posterior bleeding with blood in the throat).
   - Note the site of bleeding (anterior vs. posterior, unilateral vs. bilateral).
   - Evaluate for signs of significant blood loss (e.g., pallor, tachycardia, or hypotension).
3. **Airway and Swallowing Assessment**
   - Check for blood pooling in the oropharynx or difficulty swallowing, indicating posterior epistaxis.
   - Assess for airway compromise due to blood aspiration or clot obstruction.
4. **Determine Triage Category**
   - Determine Triage Category as per Patient Triage Categories CPG.

Patient Management & Interventions

1. **Primary Survey**
   - Perform primary survey as per Universal Care CPG.
2. **Secondary Survey**
   - Conduct a focused head and neck examination to assess the nasal cavity and oropharynx.
3. **Consider Red Flags**
   - Persistent or profuse bleeding unresponsive to initial interventions.
   - Signs of hypovolemic shock (e.g., SBP <90 mmHg in adults, tachycardia, pallor).
   - Airway compromise due to posterior bleeding or clot aspiration.
   - Known coagulopathy or anticoagulant use with uncontrolled bleeding.
   - Recurrent epistaxis with history of nasal surgery or malignancy.
4. **Evaluate for Signs of Shock or Instability**
   - Hypotension (SBP <90 mmHg in adults, or age-specific for pediatrics), tachycardia, or poor perfusion (e.g., cool, clammy skin, delayed capillary refill >2 seconds).
   - If shock is present, initiate fluid resuscitation and prioritize bleeding control.
5. **Bleeding Control Interventions**
   - Positioning
     - If the cause of epistaxis is <u>not</u> related to trauma, position the patient upright with the head tilted forward to prevent blood pooling in the oropharynx.
     - Instruct the patient to avoid swallowing blood or blowing their nose.
   - Nasal Compression

        - Apply firm, continuous pressure to the soft part of the nose (alae nasi) for 10-15 minutes.
        - Use a gauze pad or cloth to absorb blood and protect skin.
    - Nasal Packing
        - Consider nasal packing with hemostatic agents or gauze for persistent anterior bleeding.
        - Posterior packing requires hospital management; prioritize transport if suspected.
6. **Establish IV Access**
    - Indicated for hypotension, significant blood loss, or anticipated fluid administration.
    - Consider intraosseous (IO) access if IV access cannot be obtained
7. **Fluid Resuscitation**
    - **Adults:**
        - If SBP is less than 90 mmHg or HR greater than 120 BPM:
            - Consider 500 mL bolus of Normal Saline or Lactated Ringers IV/IO.
            - If SBP greater than 90 mmHg and heart rate less than 120 BPM, no IV fluids are required.
            - Repeat 500mL IV fluid bolus' as needed to maintain a SPB of 90mmHg
            - Maximum total IV/IO fluid bolus 2000mL.
    - **Pediatrics:**
        - If tachycardic for age, AND signs of poor perfusion (hypotension for age, greater than 2-second capillary refill, altered mental status, hypoxia, weak pulses, pallor, or mottled/cool skin):
            - Administer 20 mL/kg IV/IO crystalloid bolus and reassess.
            - Repeat 20 mL/kg IV/IO fluid bolus' as needed if signs of shock remain.
            - Maximum total IV/IO fluid bolus of 60mL/kg.
        - Fluid administration target is normal blood pressure for patient age.
8. **Consider ALS back-up**
    - Consider the requirement for ALS back-up as per Universal Care CPG.
    - Transport should be prioritized for uncontrolled bleeding. Do not wait on scene for ALS, meet the ALS unit en route to hospital if ALS back-up is required.
9. **Patient Reassessment**
    - Frequently Reassess the patient as per Universal Care CPG.
    - Specific Patient Reassessment Points:
        - Bleeding:
            - Reassess bleeding control every 5-10 minutes during interventions.
                - If bleeding continues maintain direct pressure.
                - Monitor for signs of airway compromise or worsening blood loss.

Patient Safety Considerations

- Avoid delays in transport for patients with red flags (e.g., posterior epistaxis, significant airway).

Clinical Insights & Additional Guidance

- Epistaxis with uncontrolled bleeding may result in severe hypovolemic shock. When initial attempts to stop bleeding fail, prioritize rapid transfer to hospital.

Documentation

- Identification: Full patient name, age, and gender.
- Mechanism / Medical Complaint: Nature and onset of bleeding, associated symptoms (e.g., dizziness, pallor).
- Injuries / Information: History of epistaxis, hypertension, trauma, anticoagulant use, or nasal surgery.
- Signs: Vital signs, site of bleeding (anterior/posterior), evidence of airway compromise or blood in oropharynx.
- Treatment:
    - Patient positioning and nasal compression performed.
    - Nasal packing used: type and response.
    - IV fluids given: type, volume, and reassessment findings.
    - ALS backup or hospital transport arranged.

- Allergies: Known allergies, especially to hemostatic agents or packing materials.
- Medications: Document use of anticoagulants, antiplatelets, or nasal sprays.
- Background: Relevant history including prior episodes, hypertension, or bleeding disorders.
- Other:
    - Scene conditions and estimated blood loss.
    - Reassessment of bleeding and airway status documented.

Key Performance Indicators

1. Document all bleeding control measures.

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